

DFI command encode — full command-bus truth table (JESD79-5 Table 241)

One channel, DDR5, 1:1 gear, N_RANKS=2, target = rank0. CA0–CA13 = the shared CA opcode/address bus; CSn/CKE/ODT/PAR/2N/RST = command sideband. green =fixed opcode bit; yellow =field / conditional; blue =static (one value, no per-phase mux). H=1, L=0 (active-low asserted), V=valid, X=don't care. **CLK**=beat (0=first CK, 1=second; CA is SDR so 1 CK = 1 command beat).

1 2-CLK commands (CA1 = L; two CK beats)

cmd	CLK	CA0	CA1	CA2	CA3	CA4	CA5	CA6	CA7	CA8	CA9	CA10	CA11	CA12	CA13	CSn0	CSn1	CKE0	CKE1	ODT0	ODT1	PAR	2N	RST
Activate ACT	0	L	L	R0	R1	R2	R3	BA0	BA1	BG0	BG1	BG2	CID	CID	CID	L	H	H	H	L	L	P	cfg	H
	1	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	R17	H	H	H	H	L	L	P	cfg	H
Write WR	0	H	L	H	H	L	BL	BA0	BA1	BG0	BG1	BG2	CID	CID	CID	L	H	H	H	w	w	P	cfg	H
	1	V	C3	C4	C5	C6	C7	C8	C9	C10	V	AP	WRp	V	CID	H	L	H	H	w	w	P	cfg	H
Read RD	0	H	L	H	H	H	BL	BA0	BA1	BG0	BG1	BG2	CID	CID	CID	L	H	H	H	w	w	P	cfg	H
	1	C2	C3	C4	C5	C6	C7	C8	C9	C10	V	AP	V	V	CID	H	L	H	H	w	w	P	cfg	H
MR Write MRW	0	H	L	H	L	L	MR0	MR1	MR2	MR3	MR4	MR5	MR6	MR7	V	L	H	H	H	L	L	P	cfg	H
	1	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7	V	V	CW	V	V	V	H	H	H	H	L	L	P	cfg	H
MR Read MRR	0	H	L	H	L	H	MR0	MR1	MR2	MR3	MR4	MR5	MR6	MR7	V	L	H	H	H	w	w	P	cfg	H
	1	L	L	V	V	V	V	V	V	V	V	CW	V	V	V	H	L	H	H	w	w	P	cfg	H

CSn (one-cold, active-low): ck0 selects target rank (rank0=L, rank1=H). ck1: ACT/MRW =H (no-ODT 2-clk, note 11); WR/RD/MRR ck1 CSn on the *non-target* rank = L to assert its ODT (note 8). **ODT w** = asserted in the WL-aligned write window (ODT_GEN), not fixed to the command beat. **PAR** P = even parity over CA0..CA13 that beat. WR vs RD differ only at CA4 (rw picks it); **ap**=CA10 ck1.

2 1-CLK commands (CA1 = H, CA0 = H; one CK beat)

cmd	CLK	CA0	CA1	CA2	CA3	CA4	CA5	CA6	CA7	CA8	CA9	CA10	CA11	CA12	CA13	CSn0	CSn1	CKE0	CKE1	ODT0	ODT1	PAR	2N	RST
Refresh all	0	H	H	L	L	H	CID3	V	V	RIR	H	L	CID	CID	CID	L	H	H	H	L	L	P	cfg	H
REFab																								
RFM all RFMab	0	H	H	L	L	H	CID3	V	V	V	L	L	CID	CID	CID	L	H	H	H	L	L	P	cfg	H
Refresh sb	0	H	H	L	L	H	CID3	BA0	BA1	RIR	H	H	CID	CID	CID	L	H	H	H	L	L	P	cfg	H
REFsb																								
RFM sb RFMsb	0	H	H	L	L	H	CID3	BA0	BA1	V	L	H	CID	CID	CID	L	H	H	H	L	L	P	cfg	H
Precharge all	0	H	H	L	H	L	CID3	V	V	V	V	L	CID	CID	CID	L	H	H	H	L	L	P	cfg	H
PREab																								
Precharge sb	0	H	H	L	H	L	CID3	BA0	BA1	V	V	H	CID	CID	CID	L	H	H	H	L	L	P	cfg	H
PREsb																								
Precharge pb	0	H	H	L	H	H	CID3	BA0	BA1	BG0	BG1	BG2	CID	CID	CID	L	H	H	H	L	L	P	cfg	H
PREpb																								
Self-Refresh entry SRE	0	H	H	H	L	H	V	V	V	V	H	L	V	V	V	L	H	L	H	L	L	P	cfg	H
SR entry freq SREF	0	H	H	H	L	H	V	V	V	V	L	L	V	V	V	L	H	L	H	L	L	P	cfg	H
Power-Down entry PDE	0	H	H	H	L	H	V	V	V	V	V	H	ODT	V	V	L	H	L	H	H	L	P	cfg	H
MPC	0	H	H	H	H	L	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7	V	L	H	H	H	L	L	P	cfg	H
NOP	0	H	H	H	H	H	V	V	V	V	V	V	V	V	V	L	H	H	H	L	L	P	cfg	H
Power-Down exit PDX	0	H	H	H	H	H	V	V	V	V	V	V	V	V	V	L	H	H	H	L	L	P	cfg	H
Deselect DES	0	CA = X (deselect)														H	H	H	H	L	L	P	cfg	H

REF/RFM: CA0–CA4 = 11001; CA9=REF(H)/RFM(L), CA10=sb(H)/ab(L). PRE: CA0–CA3=1101; CA4=pb(H)/sb-ab(L), CA10=sb(H)/ab(L). **CKE0=L** for SRE/SREF/PDE (rank0 enters low-power; PDX drives it back H). PDE **ODT0=H** = ODT persists in power-down (note 16). NOP has CSn=L; DES = CSn all H (nobody decodes CA). Static 2N/RST = one value every phase, boot/init driven.

3 How to read a row

Each command = one row (1-CLK) or two rows (2-CLK, ck0 then ck1). Left block CA0–CA13 = the shared command/address wires; right block = the sideband. Green cells are the fixed *fingerprint* the DRAM decodes; yellow are *fields* the FIELD_MUX / generators fill; blue are *static* (no per-phase mux). Decode tree: CA1 (L=2-CLK, H=1-CLK) → CA0 (L=ACT) → CA2/CA3/CA4 (command) → latch the yellow fields. In parallel, CSn qualifies which *rank* decodes (one-cold), CKE/ODT carry power & termination state, PAR is the CA parity, 2N/RST are static. **1:1 gear**: a 2-CLK command's ck0/ck1 are two successive mc_clks; at 1:2/1:4 they would be parallel phases (see gear notes in [dfi_sideband.pdf](#)).